

Name: _____ Period: _____



Unit 5: Thermodynamics

AP Classroom Topics 6 and Partial Topic 9
Unit Map

Textbook Reference: Zumdahl *Chemistry* Chapter 6, Chapter 17

Date	Day	Learning Targets	Activities	Work on this
11/10-11	U5D1	What is the difference between an endo and exothermic reaction?	• Review for Exam 4 • Begin Thermochem Unit	Study for U4 Exam
11/12-13		What have you learned about equilibrium?	• Unit 4 Exam	
11/14-17	U5D2	What is the first law of thermodynamics?	• Review Exam 4 • First Law, q calculations	
11/18-19	U5D3	What is a heating and cooling curve?	• Review calorimetry calculations • Heating and cooling curve calculations	
11/20-21	U5D4	What is the heat of a reaction?	•	
12/1-2	U5D5	How can heat be measured in the laboratory?	• Lab: Calorimetry	
12/3-4	U5D6	How else can heat of reaction be calculated?	• Hess' Law • Big Mama • Bond energies	
12/5-8	U5D7	What is entropy? What is 'free' energy?	• Entropy • Gibb's free energy	
12/9-10	U5D8	What is a thermodynamically favorable reaction?	• Wrap up notes and practice • Review	
12/11-12	U5D9	What have you learned about thermochemistry?	• Unit 5 Exam	

Unit 5: Thermodynamics Learning Targets (CED Topics 6, 9)

Unit 5	Thermodynamics
6.1	Endo and Exothermic Processes
6.1A	Explain the relationship between experimental observations and energy changes associated with a chemical or physical transformation.
	6.1.A.1: Temperature changes in a system indicate energy changes.
	6.1.A.2: Energy changes in a system can be described as endothermic and exothermic process such as the heating or cooling of a substance, phase changes, or chemical transformations.
	6.1.A.3: When a chemical reaction occurs, the energy of the system either decreases (exothermic reaction), increases (endothermic reaction), or remains the same. For exothermic reactions, the energy lost by the reacting species (system) is gained by the surroundings, as heat transfer from or work done by the system. Likewise, for endothermic reactions, the system gains energy from the surroundings by heat transfer to or work done on the system.
	6.1.A.4: The formation of a solution may be an exothermic or endothermic process, depending on the relative strengths of intermolecular / interparticle interactions before and after the dissolution process.
6.2	Energy Diagrams
6.2A	Represent a chemical or physical transformation with an energy diagram.
	6.2.A.1: A physical or chemical process can be described with an energy diagram that shows the endothermic or exothermic nature of that process.
6.3	Heat Transfer and Thermal Equilibrium
6.3A	Explain the relationship between the transfer of thermal energy and molecular collisions.
	6.3.A.1: The particles in a warmer body have a greater average kinetic energy than those in a cooler body.
	6.3.A.2: Collisions between particles in thermal contact can result in the transfer of energy. This process is called “heat transfer,” “heat exchange,” or “transfer of energy as heat.”
	6.3.A.3: Eventually, thermal equilibrium is reached as the particles continue to collide. At thermal equilibrium, the average kinetic energy of both bodies is the same, and hence, their temperatures are the same.
6.4	Heat Capacity and Thermal Equilibrium
6.4A	Calculate the heat q absorbed or released by a system undergoing heating / cooling based on the amount of the substance, the heat capacity, and the change in temperature.
	6.4.A.1: The heating of a cool body by a warmer body is an important form of energy transfer between two systems. The amount of heat transferred between two bodies may be quantified by the heat transfer equation: EQN: $q = mc\Delta T$. Calorimetry experiments are used to measure the transfer of heat.
	6.4.A.2: The first law of thermodynamics states that energy is conserved in chemical and physical processes.
	6.4.A.3: The transfer of a given amount of thermal energy will not produce the same temperature change in equal masses of matter with differing specific heat capacities.
	6.4.A.4: Heating a system increases the energy of the system, while cooling a system decreases the energy of the system.
	6.4.A.5: The specific heat capacity of a substance and the molar heat capacity are both used in energy calculations.
	6.4.A.6: Chemical systems change their energy through three main processes: heating / cooling, phase transitions, and chemical reactions.
	6.4.A.7: In calorimetry experiments involving dissolution, temperature changes of the mixture within the calorimeter can be used to determine the direction of energy flow. If the temperature of the mixture increases, thermal energy is released by the dissolution process (exothermic). If the temperature of the mixture decreases, thermal energy is absorbed by the dissolution process (endothermic).
6.5	Energy of Phase Changes
6.5A	Explain changes in the heat q absorbed or released by a system undergoing a phase transition based on the amount of the substance in moles and the molar enthalpy of the phase transition.
	6.5.A.1: Energy must be transferred to a system to cause a substance to melt (or boil). The energy of the system therefore increases as the system undergoes a solid-to-liquid (or liquid-to-gas) phase transition. Likewise, a system releases energy when it freezes (or condenses). The energy of the system decreases as the system undergoes a liquid-to-solid (or gas-to-liquid) phase transition. The temperature of a pure substance remains constant during a phase change.
	6.5.A.2: The energy absorbed during a phase change is equal to the energy released during a complementary phase change in the opposite direction. For example, the molar enthalpy of condensation of a substance is equal to the negative of its molar enthalpy of vaporization. Similarly, the molar enthalpy of fusion can be used to calculate the energy absorbed when melting a substance and the energy released when freezing a substance.
6.6	Introduction to Enthalpy of Reaction
6.6A	Calculate the heat q absorbed or released by a system undergoing a chemical reaction in relationship to the amount of the reacting substance in moles and the molar enthalpy of reaction.
	6.6.A.1: The enthalpy change of a reaction gives the amount of heat energy released (for negative values) or absorbed (for positive values) by a chemical reaction at constant pressure.
	6.6.A.2: When the products of a reaction are at a different temperature than their surroundings, they exchange energy with the surroundings to reach thermal equilibrium. Thermal energy is transferred to the surroundings as the reactants convert to products in an exothermic reaction. Thermal energy is transferred from the surroundings as the reactants convert to products in an endothermic reaction.
	6.6.A.3: The chemical potential energy of the products of a reaction is different from that of the reactants because of the breaking and forming of bonds. The energy difference results in a change in the kinetic energy of the particles, which manifests as a temperature change.
6.7	Bond Enthalpies
6.7A	Calculate the enthalpy change of a reaction based on the average bond energies of bonds broken and formed in the reaction.
	6.7.A.1: During a chemical reaction, bonds are broken and / or formed, and these events change the potential energy of the system.
	6.7.A.2: The average energy required to break all of the bonds in the reactant molecules can be estimated by adding up the average bond energies of all the bonds in the reactant molecules. Likewise, the average energy released in forming the bonds in the product molecules can be estimated. If the energy released is greater than the energy required, the reaction is exothermic. If the energy required is greater than the energy released, the reaction is endothermic.
6.8	Energy of Formation
6.8A	Calculate the enthalpy change for a chemical or physical process based on the standard enthalpies of formation.
	6.8.A.1: Tables of standard enthalpies of formation can be used to calculate the standard enthalpies of reactions. EQN: $\Delta H^\circ_{\text{reaction}} = \sum \Delta H^\circ_{\text{products}} - \sum \Delta H^\circ_{\text{reactants}}$
6.9	Hess' Law
6.9A	Represent a chemical or physical process as a sequence of steps.
	6.9.A.1: Many processes can be broken down into a series of steps. Each step in the series has its own energy change.
6.9B	Explain the relationship between the enthalpy of a chemical or physical process and the sum of the enthalpies of the individual steps.
	6.9.B.1: Because total energy is conserved (first law of thermodynamics), and each individual reaction in a sequence transfers thermal energy to or from the surroundings, the net thermal energy transferred in the sequence will be equal to the sum of the thermal energy transfers in each of the steps. These thermal energy transfers are the result of potential energy changes among the species in the reaction sequence; thus, at constant pressure, the enthalpy change of the overall process is equal to the sum of the enthalpy changes of the individual steps.

	6.9.B.2: The following are essential principles of Hess's law:
	i. When a reaction is reversed, the enthalpy change stays constant in magnitude but becomes reversed in mathematical sign. ii. When a reaction is multiplied by a factor c, the enthalpy change is multiplied by the same factor c. iii. When two (or more) reactions are added to obtain an overall reaction, the individual enthalpy changes of each reaction are added to obtain the net enthalpy change of the overall reaction.
9.1	Introduction to Entropy
9.1A	Identify the sign and relative magnitude of the entropy change associated with chemical or physical processes.
	9.1.A.1: Entropy increases when matter becomes more dispersed. For example, the phase change from solid to liquid or from liquid to gas results in a dispersal of matter as the individual particles become freer to move and generally occupy a larger volume. Similarly, for a gas, the entropy increases when there is an increase in volume (at constant temperature), and the gas molecules are able to move within a larger space. For reactions involving gas-phase reactants or products, the entropy generally increases when the total number of moles of gas-phase products is greater than the total number or moles of gas-phase reactants.
	9.1.A.2: Entropy increases when energy is dispersed. According to kinetic molecular theory (KMT), the distribution of kinetic energy among the particles of a gas broadens as the temperature increases. As a result, the entropy of the system increases with an increase in temperature.
9.2	Absolute Entropy and Entropy Change
9.2A	Calculate the standard entropy change for a chemical or physical process based on the absolute entropies (standard molar entropies) of the species involved in the process.
	9.2.A.1: The entropy change for a process can be calculated from the absolute entropies of the species involved before and after the process occurs. EQN: $\Delta S^\circ \text{ reaction} = \sum S^\circ \text{ products} - \sum S^\circ \text{ reactants}$
9.3	Gibbs Free Energy and Thermodynamic Favorability
9.3A	Explain whether a physical or chemical process is thermodynamically favored based on an evaluation of ΔG° .
	9.3.A.1: The Gibbs free energy change for a chemical process in which all the reactants and products are present in a standard state (as pure substances, as solutions of 1.0 M concentration, or as gases at a pressure of 1.0 atm (or 1.0 bar)) is given the symbol ΔG° .
	9.3.A.2: The standard Gibbs free energy change for a chemical or physical process is a measure of thermodynamic favorability. Historically, the term "spontaneous" has been used to describe process for which $\Delta G^\circ < 0$. The phrase "thermodynamically favored" is preferred instead so that common misunderstandings (equating "spontaneous" with "suddenly" or "without cause") can be avoided. When $\Delta G^\circ < 0$ for the process, it is said to be thermodynamically favored.
	9.3.A.3: The standard Gibbs free energy change for a physical or chemical process may also be determined from the standard Gibbs free energy of formation of the reactants and products. EQN: $\Delta G^\circ \text{ reaction} = \sum \Delta G_f^\circ \text{ products} - \sum \Delta G_f^\circ \text{ reactants}$
	9.3.A.4: In some cases, it is necessary to consider both enthalpy and entropy to determine if a process will be thermodynamically favored. The freezing of water and the dissolution of sodium nitrate are examples of such phenomena.
	9.3.A.5: Knowing the values of ΔH° and ΔS° for a process at a given temperature allows ΔG° to be calculated directly. EQN: $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$
	9.3.A.6: In general, the temperature conditions for a process to be thermodynamically favored ($\Delta G^\circ < 0$) can be predicted from the signs of ΔH° and ΔS° as shown in the table below:
9.4	Thermodynamic and Kinetic Control
9.4A	Explain, in terms of kinetics, why a thermodynamically favored reaction might not occur at a measurable rate.
	9.4.A.1: Many processes that are thermodynamically favored do not occur to any measurable extent, or they occur at extremely slow rates.
	9.4.A.2: Processes that are thermodynamically favored, but do not proceed at a measurable rate, are under "kinetic control." High activation energy is a common reason for a process to be under kinetic control. The fact that a process does not proceed at a noticeable rate does not mean that the chemical system is at equilibrium. If a process is known to be thermodynamically favored, and yet does not occur at a measurable rate, it is reasonable to conclude that the process is under kinetic control.
9.5	Free Energy and Equilibrium
9.5A	Explain whether a process is thermodynamically favored using the relationships between K, ΔG° and T.
	9.5.A.1: The phrase "thermodynamically favored" ($\Delta G^\circ < 0$) means that the products are favored at equilibrium ($K > 1$) under standard conditions.
	9.5.A.2: The equilibrium constant is related to free energy by the equations EQN: $K = e^{-\Delta G^\circ/RT}$ EQN: $\Delta G^\circ = -RT \ln K$.
	9.5.A.3: Connections between K and ΔG° can be made qualitatively through estimation. When ΔG° is near zero, the equilibrium constant will be close to 1. When ΔG° is much larger or much smaller than RT, the value of K deviates strongly from 1.
	9.5.A.4: Processes with $\Delta G^\circ < 0$ favor products (i.e., $K > 1$) and those with $\Delta G^\circ > 0$ favor reactants (i.e., $K < 1$).
9.6	Free Energy of Dissolution
9.6A	Explain the relationship between the solubility of a salt and changes in the enthalpy and entropy that occur in the dissolution process.
	9.6.A.1: The free energy change (ΔG°) for dissolution of a substance reflects a number of factors: the breaking of the intermolecular interactions that hold the solid together, the reorganization of the solvent around the dissolved species, and the interaction of the dissolved species with the solvent. It is possible to estimate the sign and relative magnitude of the enthalpic and entropic contributions to each of these factors. However, making predictions for the total change in free energy of dissolution can be challenging due to the cancellations among the free energies associated with the three factors cited.
9.7	Coupled Reactions
9.7A	Explain the relationship between external sources of energy or coupled reactions and their ability to drive thermodynamically unfavorable processes.
	9.7.A.1: An external source of energy can be used to make a thermodynamically unfavorable process occur. Examples include: i. Electrical energy to drive an electrolytic cell or charge a battery. ii. Light to drive the overall conversion of carbon dioxide to glucose in photosynthesis.
	9.7.A.2: A desired product can be formed by coupling a thermodynamically unfavorable reaction that produces that product to a favorable reaction (e.g., the conversion of ATP to ADP in biological systems). In the coupled system, the individual reactions share one or more common intermediates. The sum of the individual reactions produces an overall reaction that achieves the desired outcome and has $\Delta G^\circ < 0$.

AP Chem Thermodynamics Notes

6.1 Endo and Exothermic Reactions

6.2 Energy Diagrams

6.3 Heat Transfer and Thermal Equilibrium

6.4 Heat Capacity and Calorimetry

Review Time!

- Energy is required to _____ or separate particles that are attracted to one another (_____ - takes to break)
- Energy is _____ or released when bonds or attractive forces are *formed* (_____ - free to form)
- EVERY chemical or physical process is some combination of _____ attractive interactions AND the _____ of attractive forces and bonds.
- Together, they determine the energy change for the reaction

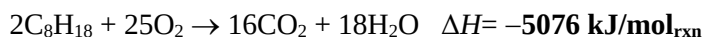
Consider the processes below. Which require energy and which free energy?

Process	
Process 1	$\text{Na}(s) \rightarrow \text{Na}(g)$
Process 2	$\text{Na}(g) \rightarrow \text{Na}^+(g) + e^-$
Process 3	$\text{F}_2(g) \rightarrow 2\text{F}(g)$
Process 4	$\text{F}(g) + e^- \rightarrow \text{F}^-(g)$

- What do we call processes that require energy? _____
- What do we call processes that release energy? _____

Exothermic- reaction which has an _____

- energy flows out of the system (more _____ than _____)
- _____ energy is changed to _____ energy
- products have _____ potential energy than reactants



Endothermic- reaction which overall _____

- energy flows into the system (More _____ than _____)
- _____ energy is changed into _____ energy
- products have higher PE than reactants

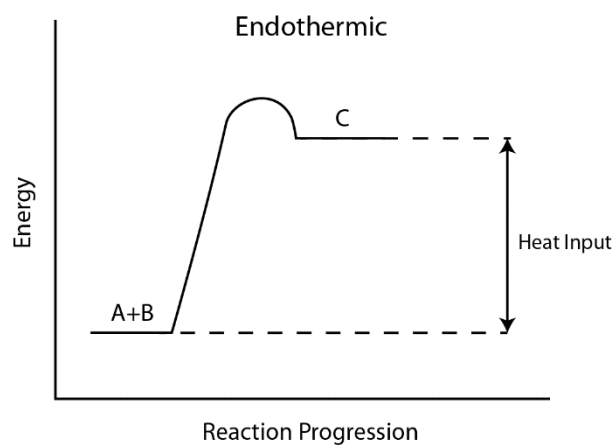
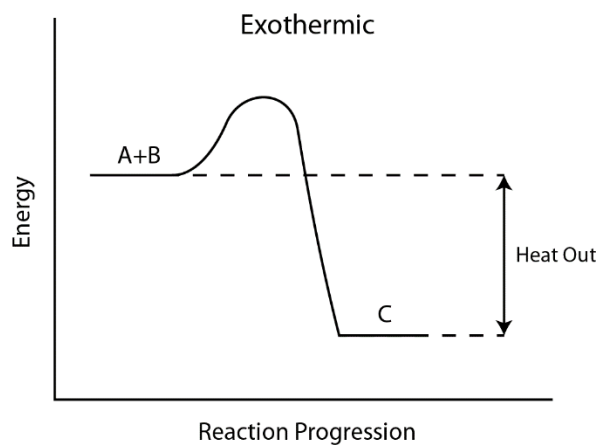


~OR~



In terms of heat, exothermic reactions are _____

In terms of heat, endothermic reactions are _____



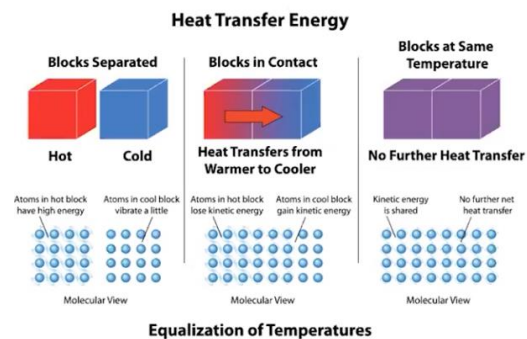
- **Energy**- the capacity to do _____ or to produce _____
- **1st Law of Thermodynamics**: Law of _____ of _____
- Energy can be converted from one form to another but it can be neither created nor destroyed. The total amount of energy in the universe is _____.

Energy Changes in Physical Processes

- **Heat** is the _____ of _____ between two bodies. Also called _____ and has the symbol _____.
- **Temperature** is a measure of the _____ energy on a _____ scale. Chemists think of temperature as _____
- **Thermal Energy $\neq$ Temperature**

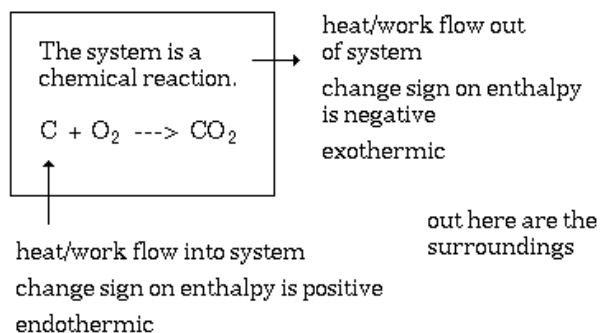
Heat Transfer

- _____ (q) occurs through molecular _____ which transfer KE (kinetic energy) from higher T molecules to lower T molecules
- When all particles have the same average KE (_____), _____ is reached
- For exothermic rxns, energy (q) is transferred _____, and _____
- For endothermic rxns, energy (q) is transferred _____, and _____



Understanding the Universe

- The _____ is our reaction (or process).
- The _____ are everything else.
- Together they make up the _____.
- Energy is _____ from one to another- either system to surroundings (_____) or surroundings to system (_____)
- Thermodynamic quantities always consist of a number and a sign (+ or -). The sign represents the _____ point of view.
- The _____ is our reaction. The _____ are everything else.



Heat _____ (either Released or Absorbed)

- The heat released or absorbed by a process (q) can be calculated: $q = mC\Delta T$
- $q =$
- $m =$
- $C =$
- $\Delta T = (T_f - T_i) =$ change in temperature- always *final - initial*
- The heat change of the system is _____ and _____ to the heat change of the surroundings

Specific Heat Capacity

- $C =$ J / (g) °C
- specific heat capacity of H₂O is _____ J/g °C
- _____ heat capacities = substances that do not conduct heat well but do hold higher heat per gram, _____ ΔT per J (_____)
- _____ heat capacities = substances that do conduct heat well and holds lower heat per gram, _____ ΔT per J (_____)

Practice: Calculate the amount of energy needed to raise 35 g of water from 25°C to 37°C

Practice: How much heat is given off when an 869 g iron bar cools from 94.0°C to 5.0°C ($C = 0.444$ J/g°C)?

First Law of Thermodynamics: Conservation of energy

- Energy can neither be created nor destroyed, so the TOTAL energy change in the UNIVERSE for any process is ZERO!

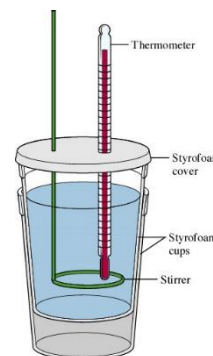
$$q(\text{universe}) = q(\text{system}) + q(\text{surroundings}) = 0$$

$$q(\text{system}) + q(\text{surroundings}) = 0$$

$$q(\text{system}) = -q(\text{surroundings})$$

Calorimetry

- _____ - insulated container of fluid (water) used for measuring the amount of _____ absorbed or released in a process.
- Energy released in a process (_____) is absorbed by the water (_____) and vice versa.
- Careful measurement of temperature changes and mass of water allow us to calculate q , energy released or absorbed.



Practice: Practice: A hot (81°C) piece of metal weighing 12 grams is dropped in a calorimeter holding 222 ml water. The water's temperature rises from 22.0°C to 24.5°C. The specific heat (C) of water is 4.184 J/g°C.

- What is the system? The surroundings?
- How much energy did the water absorb?
- What is the specific heat capacity of the metal?

Multiple Choice Practice

- How much heat is required to raise the temperature of 100. g of Fe_2O_3 from 5.0°C to 25.0°C? The specific heat of Fe_2O_3 is 0.634 J/g°C.
 - 1.27 kJ
 - 0.0634 kJ
 - 1.58 kJ
 - 0.845 kJ
- For an experiment, 50.0 g of H_2O was added to a coffee-cup calorimeter. The initial temperature of the H_2O was 22.0°C, and it absorbed 300.J of heat from an object that was carefully placed inside the calorimeter. Assuming no heat is transferred to the surroundings, which of the following was the approximate temperature of the H_2O after thermal equilibrium was reached? Assume that the specific heat capacity of H_2O is 4.2 J/(g·K).
 - 21.3°C
 - 22.0°C
 - 22.7°C
 - 23.4°C
- A 50 g sample of a metal is heated to 100°C and then placed in a calorimeter containing 100.0 g of water ($c = 4.18\text{J/g}^\circ\text{C}$) at 20°C. The final temperature of the water is 24°C. Which metal was used?
 - Lead ($c = 0.14\text{ J/g}^\circ\text{C}$)
 - Copper ($c = 0.20\text{ J/g}^\circ\text{C}$)
 - Iron ($c = 0.45\text{ J/g}^\circ\text{C}$)
 - Aluminum ($c = 0.89\text{ J/g}^\circ\text{C}$)

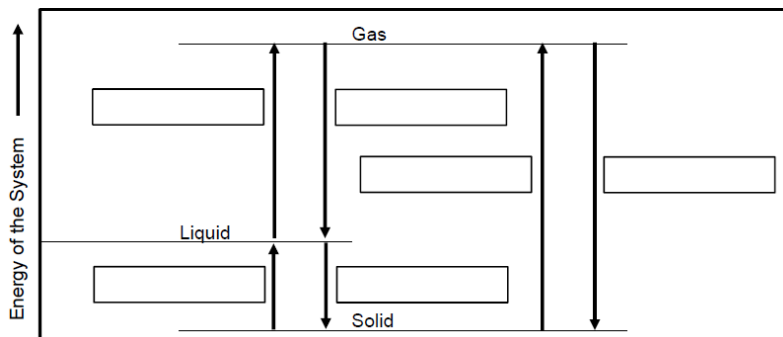
6.5 Energy of Phase Change

6.6 Intro to Enthalpy of Rxn

Heat Transfer and Phase Changes

Label the diagram with the phase changes represented for each segment:

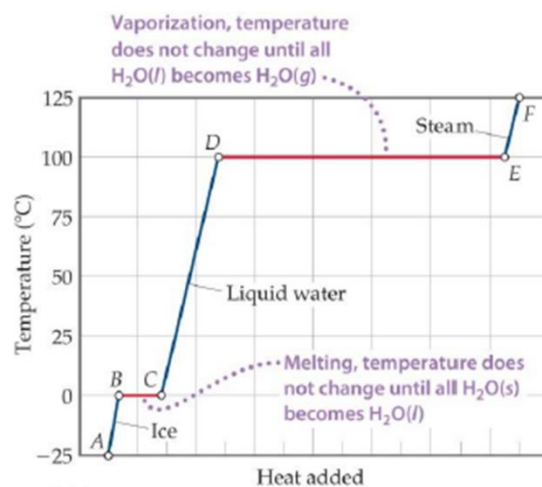
- Which phase changes are endothermic? Which are endothermic?
- Define enthalpy of fusion and enthalpy of vaporization.



- Explain why ΔH_{vap} tends to be larger than ΔH_{fus} .
- Heat of Vaporization = ΔH_{vap} = the heat required to change one mole of a substance from a _____ to a _____; units = Joules/mole = J/mol OR kJ/mol
- $n =$ _____
- How do enthalpy of boiling and condensing compare:
 $\Delta H_{\text{boil}} = + \Delta H_{\text{vap}} \cdot n$ $\Delta H_{\text{condens}} = - \Delta H_{\text{vap}} \cdot n$ _____ and _____
- Heat of Fusion = ΔH_f = the heat required to change one mole of a substance from a _____ to a _____; units = Joules/mole = J/mol OR kJ/mol
- How do H of melting and freezing compare:
 $\Delta H_{\text{melt}} = + \Delta H_f \cdot n$ $\Delta H_{\text{freeze}} = - \Delta H_f \cdot n$ _____ and _____

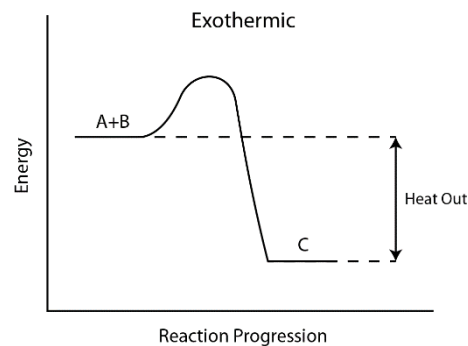
Practice: What is the energy change in the freezing of 75.0 grams of tin ($H_f = 7.03 \text{ kJ/mol}$)?

- Identify the segments of this heating curve that would involve the equation: $q = m\Delta T$
- Identify the segments of this heating curve that would involve the equation: $\Delta H = \Delta H_{\text{phase}} n$
- How much heat is needed to raise the temperature of 32.0g of water from 9°C to 104°C ? [$C_p(\text{ice}) = 2.06 \text{ J/g}^\circ\text{C}$, $C_p(\text{water}) = 4.18 \text{ J/g}^\circ\text{C}$, $C_p(\text{steam}) = 1.87 \text{ J/g}^\circ\text{C}$, $H_f(\text{water}) = 6.01 \text{ kJ/mol}$, $H_v(\text{water}) = 40.7 \text{ kJ/mol}$]



Reaction Enthalpy

- **Reaction Enthalpy or Heat of Reaction (ΔH_{rxn})** is used to _____ the _____ into (or out of) a system in a process that occurs at constant pressure.
- $\Delta H_{rxn} = \Delta H (\text{products}) - \Delta H (\text{reactants})$
- Label on the reaction energy diagram below.
- Heat of Reaction can be measured with a _____



Practice Problem: When a 1.5 g sample of KCl dissolves in 100.0 g of water in a calorimeter, the temperature drops from 22.1 °C to 21.6 °C.

- Write an equation for the process.
- Identify the system and the surroundings. (remember these are processes)
- Calculate q , the heat transferred in the process.
- Calculate ΔH_{rxn} for the dissolving process.

Molar Enthalpy (ΔH_{molrxn})

- Calorimetry is the _____ **method** of finding q , or heat transferred, for a chemical reaction, the heat involved in a physical change, or the specific heat of a substance.
- q is an _____ that depends on the _____ of substance used in the reaction. It cannot be compared unless standardized to a specific amount.... one _____.
- Molar enthalpy = heat per mole of reactant
- Can be calculated:

$$\Delta H_{molrxn} = q_{sys} / \text{moles of reactant}$$

- Why is it useful?
- Comparable because it is an _____ property
- _____ values can be applied to any amount

Practice: When a 1.5 g sample of KCl dissolves in 100.0 g of water in a calorimeter, the temperature drops from 22.1 °C to 21.6 °C.

- Write a chemical equation for the process.
- Identify the system and the surroundings. (remember these are processes)
- Calculate q , the heat transferred in the process.
- Calculate ΔH_{rxn} for the dissolving process.
- Calculate the molar heat of solvation for KCl.

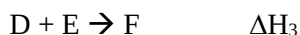
Practice: For the reaction $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$, $\Delta H = -368 \text{ kJ/mol}_{\text{rxn}}$
Calculate the heat change that occurs when 3.5 g of Na reacts with excess water.

Practice: 4.00g of ammonium nitrate are added to 100.0 mL of water in a polystyrene cup. The water in the cup is initially at a temperature of 22.5°C and decreases to a temperature of 19.3°C. Determine the heat transferred in this reaction, then determine the enthalpy of solution of ammonium nitrate in kJ/mol. Assume that the heat absorbed or released by the calorimeter is negligible.

6.9 Hess' Law

Enthalpy and Hess' Law

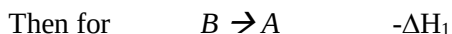
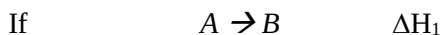
- Enthalpy of reaction is a _____: the change in enthalpy for any chemical reaction is _____, whether the reaction occurs in one step, or a series of steps.
- Hess' Law: The _____ enthalpy change in a reaction is equal to the _____ of the enthalpy changes for the _____ in the process.
- In other words, if you perform the following series of reaction steps, with the associated heats of reaction:



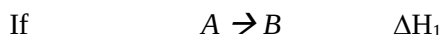
- The _____ of the heat from the collection of processes is _____ as the heat from the single step reaction
- For $A + B \rightarrow F$, $\Delta H_{\text{total}} = \Delta H_1 + \Delta H_2 + \Delta H_3$
- Helpful if ΔH_1 , ΔH_2 , ΔH_3 are already _____ values

Rules for applying Hess' Law

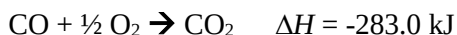
- If a reaction is _____, the sign of ΔH is _____.



- If the coefficients in a reaction are multiplied by an _____, the value of ΔH is multiplied by the same _____.



Practice: Find the heat of reaction for: $C + \frac{1}{2} O_2 \rightarrow CO$ Given:



Practice: Given the following reactions and their enthalpy changes, calculate ΔH for the reaction: $2C + H_2 \rightarrow C_2H_2$.



6.7 Bond Enthalpies

6.8 Energy of Formation (Big MAMA)

Tabulated Enthalpy Values are true at the Standard State

- ΔH° (delta H 'naught') – naught means 'at _____ state'
- standard state allows for true comparison of values while keeping other conditions _____ (standard)
- standard temperature is _____
- for gases, pressure is _____
- for a substance in solution, the concentration is _____
- for a _____ substance in a _____ state (liquid or solid), the standard state is the pure liquid or solid (not a mixture).
- for an _____, the standard state is the _____ in which the element exists under conditions of 1 atm and 25 °C.

Standard Heats of Formation Calculation- BIG MAMA

- Standard enthalpy of formation, ΔH°_f - heat released or absorbed when _____ of a compound is _____ by combination of its _____.
- Every compound has a ΔH_f . It is the amount of heat released or required to create that compound out of its base elements.

Examples:



- Because of Hess' Law, any reaction can be calculated by this sum:

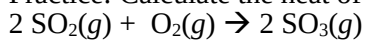
$$\Delta H_{\text{rxn}} = \Sigma \Delta H^\circ_f(\text{products}) - \Sigma \Delta H^\circ_f(\text{reactants})$$

- (Σ means "_____")
- Stuff you break apart (reactants) $\rightarrow$ Stuff you make (products)
- So in words, this equation says: The amount of heat for a reaction (ΔH_{rxn}) is equal to the heat it takes to MAKE stuff (products) *minus* the heat it takes to break stuff up (reactants).

Rules for using BIG MAMA:

- Pay attention to _____ in your equation- if 2 moles of a compound are formed, the ΔH_f must be multiplied by 2!
- The _____ matters! Make sure you keep +'s positive and -'s negative. The more negative the value of ΔH°_f , the more stable the compound.
- Note that ΔH_f for free _____ is zero (0.00 kJ/mol). (Examples: C, Na, O₂, Cl₂)

Practice: Calculate the heat of reaction of sulfur trioxide:



where: $\Delta H^\circ_f \text{SO}_2(g) = -296.8 \text{ kJ/mol}$; $\Delta H^\circ_f \text{O}_2(g) = 0.00 \text{ kJ/mol}$; $\Delta H^\circ_f \text{SO}_3(g) = -395.7 \text{ kJ/mol}$

Practice: Calculate the molar enthalpy of reaction: $\text{C}_3\text{H}_8(g) + 5\text{O}_2(g) \rightarrow 3\text{CO}_2(g) + 4\text{H}_2\text{O}(l)$

Let's Practice

1. For each reaction below, fill in the box with a subscript that specifies the type of reaction:

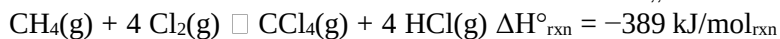


2. Circle each of the following which will have a standard heat of formation equal to zero.



3. Write a balanced molecular equation representing the enthalpy for standard heat of formation reaction of ethanol, $\text{C}_2\text{H}_5\text{OH}(l)$.

4. Use the information provided and the balanced equation to determine ΔH°_f of carbon tetrachloride.

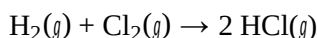


Substance	ΔH°_f
$\text{CH}_4(g)$	$-75 \text{ kJ/mol}_{\text{rxn}}$
$\text{HCl}(g)$	$-92 \text{ kJ/mol}_{\text{rxn}}$

5. Use the data regarding the standard enthalpies of formation to calculate $\Delta H^\circ_{\text{comb}}$ for the following reaction: 11

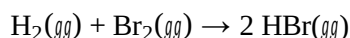


Substance	$\Delta H^\circ_f(\text{kJ/mol})$
$\text{C}_3\text{H}_6(\text{g})$	20.9 kJ/mol
$\text{CO}_2(\text{g})$	-393.5 kJ/mol
$\text{H}_2\text{O}(\text{l})$	-286 kJ/mol



6. Gaseous hydrogen and chlorine combine in the reaction above to form hydrogen chloride with an enthalpy change of -460 kJ/mol. What is the value of the heat of formation of HCl(g)?
- a. -9,20 kJ/mol b. -230 kJ/mol c. +230 kJ/mol d. +460 kJ/mol

7. If the standard enthalpies of formation of HBr(g) and $\text{Br}_2(\text{g})$ are -36 kJ mol^{-1} and $+31 \text{ kJ mol}^{-1}$ (at 298 K) respectively, what is $\Delta H^\circ_{\text{rxn}}$ for the following reaction?

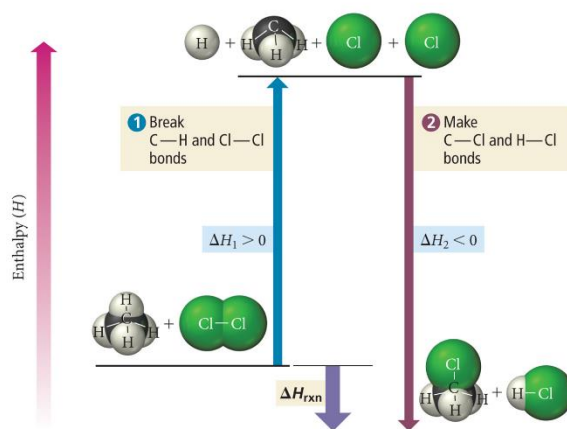


- a. -103 kJ mol^{-1} b. -67 kJ mol^{-1} c. $+67 \text{ kJ mol}^{-1}$ d. $+103 \text{ kJ mol}^{-1}$

Enthalpy of Reaction from Bond Energies

- Chemical reactions involve a combination of _____ bonds (always _____ thermic- TTB) and _____ new bonds (always _____ thermic- FTF).
- The $\Delta H^\circ_{\text{reaction}}$ can be _____ by comparing the cost of breaking old bonds to the income from making new bonds.
- Bond Energy-** The amount of energy it takes to _____ one mole of a specific bond in a compound.
 - Usually measured in the gas state
 - Bond breaking is always _____, so bond energies are _____ values
 - The actual bond energy depends on the surrounding atoms and other factors, so **average bond energies** to estimate the ΔH_{rxn} .
- Because bond energies are averages, $\Delta H^\circ_{\text{rxn}}$ is an _____.
- Works best when all reactants and products in _____
- Bond breaking is _____ thermic, $\Delta H(\text{breaking}) > 0$.
- Bond making is _____ thermic, $\Delta H(\text{making}) < 0$.

Estimating the Enthalpy Change of a Reaction from Bond Energies



$$\Delta H_{\text{rxn}} = \sum (\Delta H(\text{bonds broken})) - \sum (\Delta H(\text{bonds formed}))$$

Calculating ΔH_{rxn} from Bond Energies

Hydrogen gas, a potential fuel, can be made by the reaction of methane gas and steam.



Use bond energies to calculate ΔH_{rxn} for this reaction.

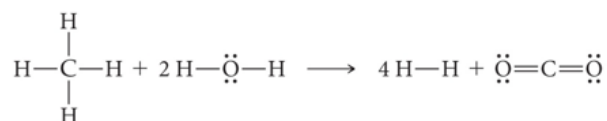


Table 8.4 ► Average Bond Energies (kJ/mol)

Single Bonds						Multiple Bonds	
H—H	432	N—H	391	I—I	149	C=C	614
H—F	565	N—N	160	I—Cl	208	C≡C	839
H—Cl	427	N—F	272	I—Br	175	O=O	495
H—Br	363	N—Cl	200			C=O*	745
H—I	295	N—Br	243	S—H	347	C≡O	1072
		N—O	201	S—F	327	N=O	607
C—H	413	O—H	467	S—Cl	253	N=N	418
C—C	347	O—O	146	S—Br	218	N≡N	941
C—N	305	O—F	190	S—S	266	C≡N	891
C—O	358	O—Cl	203			C=N	615
C—F	?	O—I	234	Si—Si	340		
C—Cl	339			Si—H	393		
C—Br	276	F—F	154	Si—C	360		
C—I	240	F—Cl	253	Si—O	452		
C—S	259	F—Br	237				
		Cl—Cl	239				
		Cl—Br	218				
		Br—Br	193				

*C=O(CO₂) = 799

Study Guide or: How I Learned to Stop Worrying and Love Thermochem

First: determine what you're being asked to find: ΔH_{rxn} or something else?

1. Asked to find ΔH_{rxn} ? 3 Options!

Given ?	Asked to find?	Use:
1. A lot of bond energies 2. A reaction without ΔH_{rxn}	ΔH_{rxn}	1. Draw the Lewis structures. 2. Use the following: $\Delta H_{\text{rxn}} = \Sigma(\text{BE}_{\text{reactants}}) - \Sigma(\text{BE}_{\text{products}})$
1. A lot of heats of formation (ΔH_f) 2. A reaction without ΔH_{rxn}	ΔH_{rxn}	$\Delta H_{\text{rxn}} = \Sigma \Delta H_f (\text{products}) - \Sigma \Delta H_f (\text{reactants})$
1. Multiple reactions with ΔH 2. A goal reaction without ΔH_{rxn}	ΔH_{rxn}	Hess's Law! Rearrange the equations to make the goal equation, then combine your new ΔH 's (remember, what you do to an equation you must do to ΔH !)
1. info to calculate q (using $mC\Delta T$ or $nC\Delta T$) 2. Moles/grams of a chemical	ΔH_{rxn}	$\Delta H_{\text{rxn}} = \frac{q}{\text{mol}_{\text{rxn}}}$

2. Asked to find something else? 3 Options!

Given ?	Asked to find?	Use:
1. A reaction with ΔH_{rxn} 2. Either: a. g or mol of a substance b. energy change (J or kJ)	1. Either: a. g or mol of a substance b. energy change (J or kJ)	Stoich! Don't forget to convert between moles of your substance and $\text{moles}_{\text{rxn}}$
1. A phase change (vaporizing, condensing, freezing or melting) 2. ΔH_{vap} or ΔH_{fus}	Energy change (heat absorbed or released)	$q = n\Delta H$
1. A temperature change—— 2. Mass or moles of a substance 3. Heat capacity (J/g°C)	Energy change (heat absorbed or released)	$q = mC\Delta T$

9.1 Intro to Entropy
9.2 Absolute Entropy and Entropy Change
9.3 Gibbs Free Energy and Thermodynamic Favorability
9.4 Thermodynamic and Kinetic Control

Thermodynamic Favorability or “Spontaneity”

- Processes are determined to be “_____” (spontaneous) if they occur without outside intervention (such as being coupled to an outside energy source)
- may be fast or slow- NOT determined by _____
- depends on two factors- _____ (ΔH) and _____ (ΔS)

Entropy, ΔS

- the number of “_____” in which a system can transfer energy
- (kind of) measure of randomness or disorder
- associated with probability (There are more ways for something to be disorganized than organized).
- for a reaction, the _____ in entropy (ΔS) from products to reactants is very important.
- an _____ in entropy is thermodynamically _____ for a reaction

2nd Law of Thermodynamics

- In any spontaneous process there is always an _____ in the entropy of the universe.
- The energy of the universe is _____ but the entropy of the universe is _____.
- Therefore a process that shows an _____ in entropy ($\Delta S > 0$) is favored by nature.

3rd Law of Thermodynamics

- As a system approaches 0 Kelvin (absolute zero), the entropy of that system approaches zero.

Change in Entropy, ΔS

- For a process:
- _____ in entropy is thermodynamically _____ and
- _____ in entropy is thermodynamically _____.

$\Delta S > 0$, FAVORABLE (+ _____!)

$\Delta S < 0$, UNFAVORABLE(- _____!)

- Notice that this is opposite sign convention from ΔH

($-\Delta H$ _____ and $+\Delta H$ _____!)

Predicting Change in Entropy, ΔS

- $A(s) \rightarrow A(l)$ Entropy _____ going from a solid to a liquid to a gas.
- $MX(s) \rightarrow MX(aq)$ Entropy _____ when solutions are formed.
- $Q(290\text{ K}) \rightarrow Q(350\text{ K})$ Entropy _____ with increasing temperature
- $2H_2O(g) \rightarrow 2H_2(g) + O_2(g)$ Entropy usually _____ in a reactions when more moles of products are formed. Exception:

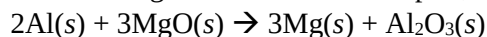
Calculating ΔS

- ΔS can be calculated using a formation equation much like “_____”

$$\Delta S^{\circ}_{\text{reaction}} = \Sigma S^{\circ}_{\text{prod}} - \Sigma S^{\circ}_{\text{react}}$$

- Important: ΔS is _____ for elements
- ΔS values are much _____ than ΔS values and are tabulated in _____ /mol K instead of _____ /mol K. This will be important later!

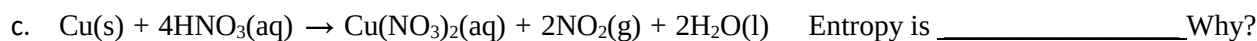
Practice: Given the following standard molar entropies, calculate ΔS° for the reaction:



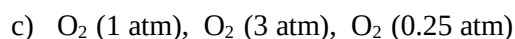
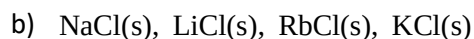
Substance	ΔS° value (J/mol K)
Al(s)	28.0 J/mol K
MgO(s)	27.0 J/mol
Mg(s)	33.0 J/mol K
Al ₂ O ₃ (s)	51.0 J/K

Practice:

1) For the following reactions, is the entropy of the reaction *increasing* or *decreasing*?



2) Place the following in order of increasing entropy:



Thermodynamic Favorability and Free energy, ΔG

Thermodynamic Favorability is the combination of TWO factors:

_____ ($\Delta H < 0$ is favorable) and

_____ ($\Delta S > 0$ is favorable)

$$\Delta G = \Delta H - T\Delta S$$

- ΔG , or _____ - Free energy change is a measure of the thermodynamic favorability (spontaneity) of a reaction. It is the maximum _____ available from the system.
- Determining the thermodynamic favorability of a reaction considers both enthalpy and entropy
- This is called the Gibbs-Helmholtz equation... or the Big Grampa Eqn
- ***When working this, change ΔS from J to kJ***

Thermodynamic Favorability (Spontaneity) is determined by the SIGN of ΔG

- A spontaneous (thermodynamically favored) reaction has a _____ ΔG ($\Delta G < 0$).
- A nonspontaneous (thermodynamically NOT favored) reaction has a _____ ΔG ($\Delta G > 0$).
- A reaction at _____ (forward and backward _____ favored) has a $\Delta G = 0$.
- _____ is an example of a system at equilibrium- For example, when ice melts, $\Delta H > 0$ (endothermic), $\Delta S > 0$ and $\Delta G = 0$ at 0°C .
- _____ - a reaction that has a $\Delta G < 0$ is thermodynamically favorable, but the _____ is not determined by ΔG . This means that it may proceed very slowly, like rusting of metal, or some radioactive decay. Both are spontaneous, but slow because they are under 'kinetic control'

Predicting thermodynamic favorability using $\Delta G = \Delta H - T\Delta S$

ΔH	ΔS	ΔG	Is the Reaction Spontaneous?

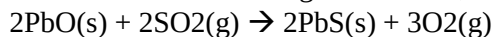
What does spontaneous at all temperatures mean?

What does spontaneous at no temperatures mean?

What does spontaneous at high temperatures mean?

What does spontaneous at low temperatures mean?

Practice: For the given reaction and the following information, calculate ΔG° at 25 °C in kJ/mol.



	ΔH° (kJ/mol)	ΔS° (J/molK)
PbO(s)	-218.0	70.0
SO ₂ (g)	-297.0	248.0
PbS(s)	-100.0	91.0
O ₂ (g)	-----	205.0

- ΔG° can also be calculated using a Big Mama equation.

$$\Delta G^\circ = \sum \Delta G_f^\circ (\text{products}) - \sum \Delta G_f^\circ (\text{reactants})$$

- Note: ΔG_f° for a free element in its standard state is _____ and these values are in _____

Practice: Ex. Given the equation: $\text{N}_2\text{O}_4(g) \rightarrow 2\text{NO}_2(g)$
and the following data, calculate ΔG° .

$$\Delta G_f^\circ \text{ for } \text{N}_2\text{O}_4(g) = 97.82 \text{ kJ/mol}$$

$$\Delta G_f^\circ \text{ for } \text{NO}_2(g) = 51.30 \text{ kJ/mol}$$

When $\Delta G = 0$, system is a equilibrium

- The equilibrium point occurs at the lowest value of free energy available to the reaction system.

$$\Delta G = G_{\text{prod}} - G_{\text{react}} = 0$$

***The most common example is a _____ !

Ex. Given for the reaction: $\text{Hg}(l) \rightarrow \text{Hg}(g)$

$\Delta H^\circ = 61.3 \text{ kJ/mol}$ and $\Delta S^\circ = 100.0 \text{ J/molK}$, Calculate the normal boiling point of Hg.

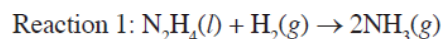
When ΔG is positive.....

A reaction where $\Delta G > 0$ will not occur independently. External sources of energy can be used to drive reactions with positive ΔG values. These are known as _____ reactions

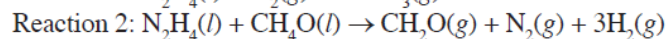
- Electricity may be used to cause a reaction to occur through electrolysis. We will look at this in our electrochemistry unit (Ch 18).
- Light may be used (photoionization or absorption of photons (as in photosynthesis))
- The reaction may be couple with another that has a very negative ΔG value, as in the conversion of ADP to ATP.

Multiple Choice Practice

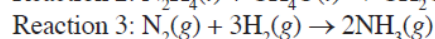
Use the following information to answer questions 4 – 6.



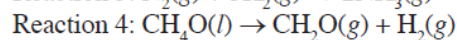
$$\Delta H = ?$$



$$\Delta H = -37 \text{ kJ/mol}_{\text{rxn}}$$

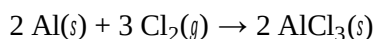


$$\Delta H = -46 \text{ kJ/mol}_{\text{rxn}}$$

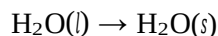


$$\Delta H = -65 \text{ kJ/mol}_{\text{rxn}}$$

- What is the enthalpy change for reaction 1?
 - $-148 \text{ kJ/mol}_{\text{rxn}}$
 - $-56 \text{ kJ/mol}_{\text{rxn}}$
 - $-18 \text{ kJ/mol}_{\text{rxn}}$
 - $+148 \text{ kJ/mol}_{\text{rxn}}$
- If reaction 2 were repeated at a higher temperature, how would the reaction's value for ΔG be affected?
 - It would become more negative because entropy is a driving force behind this reaction.
 - It would become more positive because enthalpy is a driving force behind this reaction.
 - It would become more negative because the gases will be at a higher pressure.
 - It will stay the same; temperature does not affect the value of ΔG .
- Under what conditions would reaction 3 be thermodynamically favored?
 - It is always favored.
 - It is never favored.
 - It is only favored at low temperatures.
 - It is only favored at high temperatures.



- The reaction above is not thermodynamically favored under standard conditions, but it becomes thermodynamically favored as the temperature decreases toward absolute zero. Which of the following is true?
 - ΔS and ΔH are both negative.
 - ΔS and ΔH are both positive.
 - ΔS is negative, and ΔH is positive.
 - ΔS is positive, and ΔH is negative.



- Which of the following is true for the above reaction?
 - The value of ΔS is positive.
 - The value of ΔG is positive at 298 K.
 - The value of ΔH is positive.
 - The reaction is favored at 1.0 atm and 298 K.
- A chemical reaction has an equilibrium constant, K , equal to 1.0×10^{-6} . If, at a given point in the reaction, the value for the reaction quotient Q is determined to be 2.5×10^{-8} , what is true about Gibbs's free energy at that moment?
 - $\Delta G = 0$
 - $\Delta G > 0$
 - $\Delta G < 0$
 - The value of ΔG cannot be determined.

7. In which of the following reactions is entropy increasing?

- a. $2 \text{SO}_2(g) + \text{O}_2(g) \rightarrow 2 \text{SO}_3(g)$ c. $\text{H}_2(g) + \text{Cl}_2(g) \rightarrow 2 \text{HCl}(g)$
 b. $\text{CO}(g) + \text{H}_2\text{O}(g) \rightarrow \text{H}_2(g) + \text{CO}_2(g)$ d. $2 \text{NO}_2(g) \rightarrow 2 \text{NO}(g) + \text{O}_2(g)$

Use the following information to answer questions 8-9.

When calcium chloride (CaCl_2) dissolves in water, the temperature of the water increases dramatically.

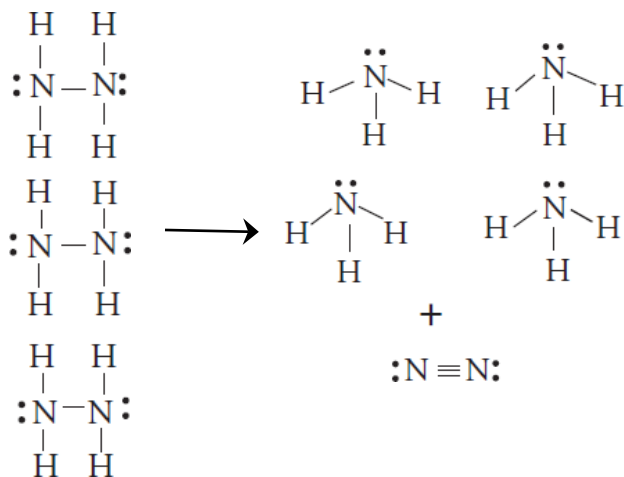
8. During this reaction, energy transfers from:

- a. the reactants to the products. c. the system to the surroundings.
 b. the reactants to the system. d. the products to the surroundings.

9. Which is the primary driving factor behind this reaction?

- a. Entropy c. Both enthalpy and entropy
 b. Enthalpy d. Neither enthalpy and entropy

10. The reaction shown in the diagram below is accompanied by a large increase in temperature. If all molecules shown are in their gaseous state, which statement accurately describes the reaction?



- a. It is an exothermic reaction in which entropy increases.
 b. It is an exothermic reaction in which entropy decreases.
 c. It is an endothermic reaction in which entropy increases.
 d. It is an endothermic reaction in which entropy decreases.

11. When solid ammonium chloride, $\text{NH}_4\text{Cl}(s)$, is added to water at 25°C , it dissolves and the temperature of the solution decreases. Which of the following is true for the values of ΔH and ΔS for the dissolving process?

- | ΔH | ΔS |
|-------------|------------|
| a. Positive | Negative |
| b. Positive | Positive |
| c. Negative | Positive |
| d. Negative | Negative |